

Business Problem Statement

Data Professional Survey Analysis

Organizations in the data industry need to understand the characteristics, compensation, skills, and job satisfaction of data professionals in order to make better workforce and career-related decisions.

The available dataset contains survey responses from data professionals across different countries, job roles, industries, and experience levels. The survey captures information such as current job title, salary range, favourite programming language, age, country, work-life balance satisfaction, salary satisfaction, and the perceived difficulty of entering the data field.

The challenge is to transform this raw survey data into meaningful insights that can help identify patterns in the data profession, including differences in salaries across job roles and demographics, the most popular programming languages, geographic differences, employee satisfaction, and the perceived difficulty of entering the data industry.

Overarching Business Question

“What insights can be derived from data professional survey responses to understand salary trends, job roles, skills, satisfaction levels, and career entry barriers across the data industry?”

Key Business Questions

The analysis aims to answer the following questions:

1. Which data-related job roles have the highest average salaries?
2. How does average salary vary across different job titles?
3. Which programming languages are most commonly preferred by data professionals?
4. How are survey respondents distributed across different countries?
5. How satisfied are data professionals with their salaries?
6. How satisfied are data professionals with their work-life balance?
7. How difficult do professionals perceive it to be to enter the data industry?
8. How does average salary differ between male and female respondents?
9. What patterns can be identified across job roles, salary, skills, demographics, and satisfaction?

Project Objectives

- Clean and transform the raw survey data into an analysis-ready dataset.
- Standardize important categorical fields such as job titles and programming language responses.
- Convert salary information into a usable numerical measure for analysis.
- Analyze salary differences across job roles and demographic groups.
- Identify the most popular programming languages among survey respondents.

- Analyze geographic distribution of survey participants.
- Measure satisfaction with salary and work-life balance.
- Understand how respondents perceive the difficulty of entering the data industry.
- Build an interactive Power BI dashboard that presents the findings clearly.

Deliverables

1. Data Preparation & Transformation

Use Power Query to clean and transform the raw survey dataset, including simplifying categorical responses and converting salary information into a usable format for analysis.

2. Data Analysis

Analyze the survey data to identify patterns in:

- Job roles
- Average salary
- Programming language preferences
- Country distribution
- Salary satisfaction
- Work-life balance
- Difficulty of entering the data industry
- Gender-based salary differences

3. Visualization & Insights

Build an interactive **Power BI dashboard** that presents the major findings through appropriate charts, cards, gauges, and other visualizations.

The dashboard should allow users to quickly understand salary patterns, job-role differences, programming language preferences, respondent demographics, and satisfaction levels.

4. Business Insights & Recommendations

Summarize the major findings from the analysis and identify meaningful observations that could help:

- Aspiring data professionals understand the current data job market.
- Professionals compare different data-related career paths.
- Organizations understand workforce characteristics and satisfaction.
- Training and education providers identify relevant skills and career barriers.

5. GitHub Portfolio Repository

Create a well-structured GitHub repository containing the project documentation, cleaned/processed dataset where appropriate, Power BI dashboard, screenshots, and supporting project files.

Expected Outcome

The final analysis will transform raw survey responses into an interactive business intelligence dashboard that provides a clear view of the data profession, particularly salary trends, job roles, programming language preferences, geographic distribution, satisfaction, and career entry difficulty.